

Large Scale Sand Simulation on Under The Boardwalk

Jonathan Davies
DNEG Animation
London, UK
jsd@dneg.com

Nicholas New
DNEG Animation
London, UK
nsn@dneg.com

Cheten Sharma
DNEG Animation
London, UK
chsh@dneg.com



Figure 1: Under The Boardwalk Artwork and Film Stills - ©Paramount Animation

ABSTRACT

In the 2023 Feature Animation **Under the Boardwalk** a Romeo and Juliet meets West Side Story themed musical comedy; we enter the miniature world of the Hermit Crab. Set on the Jersey Shore, the rivalry between the sea crabs descending on the home of the resident land crabs during spring break gave us lots of sand interaction shots. Our workflows had to handle closeup macro shots with one or two crab characters, full battle sequences with hundreds of crowd crabs, and human scale characters. With hundreds of FX shots featuring fully simulated sand interaction and many beach environments needing sculpting and shaping to match dressed assets, we had to turn to automation techniques to help us tackle this number of shots efficiently. Our **envfinishing** step was used to preprocess the shot beach geometry to add detail, deform the sand around the static set objects and add footprints. The FX team used this geometry as a basis for the grain simulations and include any water interaction with rain or waves. These caches had to inherit the large scale textural detail and the fine, per grain variations provided by the surfacing department to seamlessly blend the sand and beach geometry.

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CCS CONCEPTS

• **Computing methodologies** → **Mesh geometry models**; **Physical simulation**; *Texturing*; *Ray tracing*.

KEYWORDS

Animation, Dynamic/Simulation, Geometry, Rendering, Production

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1 INTRODUCTION

The sand was the biggest challenge on the film, featured in hundreds of shots. The 3 main challenges to solve were:

- Adding local detail to the large beach environment.
- Configuring sand simulations in a repeatable way.
- Blending the sand grains into the wider beach geometry.

We automated most of these steps in Houdini (SideFX) to give us completed shots or a great starting point for custom simulation.

2 SAND REQUIREMENTS

The early artwork for the film showed the crab's macro world, with very prominent sand grains with varying shape and material. We prepared sections of beach with the characters next to different size sand grains to get signoff on the look and feel for the film.

The grain size altered the perceived scale of the crabs and dictated our simulation and rendering approaches, with super-fine sand rendered mainly as points or coarser sand needing more geometry instances to give grain detail and variation. Luckily we settled close to the artwork style. The best example of existing work like this was the Pixar animated short film *Piper*, and this became the gold standard for what we wanted to achieve, but scaled up to a feature length production.

3 INITIAL SAND TESTS

Early in the production process, we didn't know if we could run full sand simulations for the number and variety of shots. Early tests looked at a hybrid approach to see if we could deform the ground, just simulating the boundary edges. This was promising, and a great fallback but further testing with the Houdini Grains solver and the automation plan, gave us confidence that fully simulated sand patches would be possible with time and budget constraints.

The locations in the film were densely populated with litter and debris from the human world. With large expanses of beach compared to the crab scale, we needed to sculpt the sand around the debris per shot.

The beach locations were manually sculpted to the larger buildings and we set up a pipeline step **envfinishing** to handle the sculpting around the detailed set dressed object to prep the ground for the FX Simulation stages.

4 ENVFINISHING - SAND FOOTPRINTS

The first step in this process was to add footprint trails to show the character history from animation in previous shots, or a journey implied by the story. We built a simple template scene to allow artists to draw or create curves to describe the character trails. The automated step was to preload and optimise all the set geometry and bake proxy anim caches of each character to give the artist a lightweight scene to work with. The curve drawing was quick manual step with the default curve / draw tools provided in the template scene. Curves could also be imported from other shots, or procedurally generated for crowds. The curves were tagged with metadata including character names and start / end fade values to control the footprint depth.

5 ENVFINISHING - SAND SCULPT

Next, the sand deformation around the environment geometry was calculated and the footprint curves in the last step were used to generate trails in the sand. The challenge in generating this geometry was adding iterations of subdivided detail to a low-res mesh. The detail patches were small isolated regions across the main beach surface. Most quad-based subdivision techniques were quite slow with scattered face selections to subdivide, would result in artefacts, or huge memory usage at render time when dicing. The most reliable and repeatable solution was to triangulate the geometry. Using stored edge lengths, consecutive remeshes on different or overlapping regions added resolution where needed.

6 SAND SCATTERING

When simulating the character and sand interactions we used the animation to determine the regions that the characters interacted

with. This included the depth if the crabs were burying themselves or breaking through a mound of sand. We limited the depth of the active sand due to the huge increase in grains generated and simulation complexity. The sand grains were generated using the **Grain Source** node, creating a cleanly packed volume of points. Many attributes were inherited from the source surface to help with shading and blending between the remaining ground geometry and the scattered grains. We had to be very careful when scattering and running the simulations, as we couldn't afford any settling of the sand, as this made the blend between grains and geometry very noticeable.

We aimed for 2.5m sand instances to keep the memory footprint low, working out which grains were best to render as instances. We prioritised grains that were at the top of the local sand depth, or large on screen.

7 BLENDING GRAINS AND GEOMETRY

We had to make sure there was no visible join between the sand grains and geometry. A rest attribute: **rmanP_Pref** allowed the initial position relative to the ground geo to be stored - helping with triplanar texture layers that were applied. A test patch of instanced grains was used to generate the texture layers for the beach geometry to define the material properties. The actual shot grains randomly sampled this texture to maintain a consistent look. UVs were copied from the ground plane for use when textural detail needed to be painted on the beach. The simulated sand revealed vertical streaks from the texture through the sand layers, as grains in the same X/Z space shared UV coordinates. To solve this we calculated the local grain depth, with deeper grains given more jitter to the ground sample position adding diffusion to the texture.

To help the grains and the geometry blend seamlessly, the grains were sunk into the ground towards the sim edges. The density at the edges was reduced slightly to give a gradual reveal of the underlying displaced geometry.

8 SAND SIMULATION

The sand simulation settings were split according to wetness levels. The grain simulation focused on the dry and damp sand only, with the really wet sand, being handled with geometry deformation, as this was seen infrequently.

9 ANIMATION CHALLENGES

The animators played a huge part in helping the automation approach work. Setting rules on foot / claw depth and speed, ensured a more predictable result from the simulations. We still had many hero shots that required custom FX work such as crabs burrowing into the beach or objects thrown into the sand.

10 WAVE AND RAIN INTERACTION

A wetmap attribute was stored on the static and dynamic sand grains and beach geometry to drive the wet look from the wave and rain interaction.